

## LAMEZIA TERME, Italy

WMO: 163620

Lat: 38.905N

Lon: 16.242E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 33.5       | 36.2 | 19.7                          | 14.8 | 41.7 | 24.6 | 18.7 | 43.7 | 30.9                  | 55.4 | 27.8 | 55.5 | 5.9                   | 90   | 0.447 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 16.4                   | 89.3            | 73.3 | 86.3 | 75.3 | 84.5 | 75.0 | 79.6                | 83.8 | 78.5 | 83.1 | 77.2 | 82.2 | 9.7                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 78.6                            | 148.9 | 82.4 | 77.0 | 140.9 | 81.7 | 75.4 | 133.3 | 80.8 | 43.2          | 83.9 | 41.9 | 83.6 | 40.5 | 82.2 |                   |
|                                 |       |      |      |       |      |      |       |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 25.6              | 22.4 | 19.5 | DB                         | 27.5 | 96.0               | 2.8 | 3.9                                                | 25.5 | 98.8       | 23.9 | 101.0      | 22.4 | 103.2      | 20.4 | 106.0 |
|                   |      |      | WB                         | 25.6 | 81.8               | 2.8 | 2.2                                                | 23.6 | 83.4       | 21.9 | 84.7       | 20.4 | 85.9       | 18.4 | 87.6  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 62.3   | 50.1  | 50.0  | 53.2  | 57.8  | 64.5  | 71.2  | 75.7  | 76.6  | 71.2  | 65.3  | 58.3  | 52.3 |
|                                                                               | DBStd       | 10.47  | 4.88  | 4.87  | 4.53  | 4.31  | 4.46  | 4.40  | 3.33  | 3.11  | 4.29  | 4.52  | 5.24  | 5.02 |
|                                                                               | HDD50       | 175    | 58    | 55    | 20    | 3     | 0     | 0     | 0     | 0     | 0     | 5     | 34    |      |
|                                                                               | HDD65       | 2188   | 461   | 419   | 367   | 220   | 63    | 4     | 0     | 4     | 50    | 205   | 395   |      |
|                                                                               | CDD50       | 4651   | 62    | 56    | 118   | 236   | 450   | 637   | 798   | 824   | 636   | 474   | 256   | 104  |
|                                                                               | CDD65       | 1190   | 0     | 0     | 0     | 4     | 48    | 192   | 333   | 359   | 190   | 59    | 5     | 0    |
|                                                                               | CDH74       | 9439   | 0     | 0     | 7     | 26    | 240   | 1294  | 2993  | 3400  | 1279  | 194   | 6     | 0    |
|                                                                               | CDH80       | 2329   | 0     | 0     | 1     | 2     | 52    | 292   | 727   | 992   | 247   | 16    | 0     | 0    |
| Wind                                                                          | WSAvg       | 8.6    | 9.2   | 9.1   | 9.6   | 9.8   | 8.9   | 7.7   | 7.9   | 7.3   | 7.8   | 7.9   | 8.8   | 8.8  |
| Precipitation                                                                 | PrecAvg     | 37.8   | 4.7   | 4.0   | 3.2   | 2.7   | 1.9   | 1.1   | 0.8   | 1.1   | 3.0   | 4.2   | 5.2   | 6.1  |
|                                                                               | PrecMax     | 52.8   | 7.7   | 13.9  | 5.6   | 4.9   | 3.7   | 4.6   | 3.1   | 3.5   | 5.9   | 11.8  | 8.8   | 15.0 |
|                                                                               | PrecMin     | 24.5   | 2.1   | 1.0   | 0.6   | 1.5   | 0.4   | 0.2   | 0.1   | 0.2   | 0.9   | 0.8   | 1.5   | 1.7  |
|                                                                               | PrecStd     | 7.2    | 1.7   | 2.7   | 1.5   | 0.9   | 1.0   | 0.9   | 0.7   | 1.1   | 1.3   | 2.6   | 1.8   | 3.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 64.4  | 65.9  | 72.5  | 77.4  | 86.2  | 92.8  | 92.9  | 93.7  | 89.6  | 82.0  | 74.4  | 67.8 |
|                                                                               |             | MCWB   | 56.6  | 56.3  | 59.2  | 61.2  | 67.2  | 71.2  | 72.5  | 74.9  | 72.7  | 68.6  | 64.0  | 60.1 |
|                                                                               | 2%          | DB     | 61.2  | 62.3  | 65.9  | 71.8  | 79.5  | 86.2  | 87.8  | 89.2  | 85.6  | 78.0  | 71.6  | 64.4 |
|                                                                               |             | MCWB   | 56.0  | 55.5  | 57.1  | 59.1  | 65.7  | 71.5  | 75.6  | 74.5  | 73.1  | 68.1  | 64.5  | 58.0 |
|                                                                               | 5%          | DB     | 59.6  | 59.7  | 62.8  | 68.3  | 75.7  | 82.7  | 85.7  | 86.3  | 82.5  | 75.7  | 69.5  | 62.6 |
|                                                                               |             | MCWB   | 55.0  | 54.0  | 56.4  | 58.8  | 65.7  | 72.8  | 75.6  | 76.6  | 72.4  | 67.6  | 62.9  | 57.1 |
|                                                                               | 10%         | DB     | 58.6  | 58.6  | 60.9  | 65.9  | 73.2  | 80.5  | 84.0  | 84.5  | 80.6  | 73.9  | 67.6  | 60.9 |
|                                                                               |             | MCWB   | 54.1  | 53.3  | 55.3  | 58.4  | 65.2  | 72.6  | 75.3  | 76.3  | 71.7  | 66.6  | 61.1  | 56.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 59.4  | 59.0  | 61.2  | 64.5  | 71.6  | 80.0  | 80.6  | 81.6  | 79.2  | 72.9  | 67.9  | 61.3 |
|                                                                               |             | MCDB   | 62.1  | 62.6  | 67.0  | 70.3  | 77.2  | 82.4  | 84.2  | 85.2  | 83.3  | 76.4  | 71.3  | 65.3 |
|                                                                               | 2%          | WB     | 57.6  | 56.9  | 58.9  | 62.1  | 69.5  | 77.5  | 79.0  | 80.0  | 76.3  | 70.9  | 65.5  | 59.6 |
|                                                                               |             | MCDB   | 60.6  | 60.5  | 63.8  | 67.5  | 74.8  | 81.3  | 83.4  | 84.3  | 81.3  | 75.3  | 69.8  | 63.0 |
|                                                                               | 5%          | WB     | 56.0  | 55.4  | 57.5  | 60.7  | 67.7  | 75.4  | 77.8  | 78.9  | 74.5  | 69.3  | 63.6  | 58.2 |
|                                                                               |             | MCDB   | 59.1  | 58.9  | 61.6  | 65.6  | 73.0  | 79.5  | 82.6  | 83.7  | 79.9  | 74.2  | 68.1  | 61.7 |
|                                                                               | 10%         | WB     | 54.4  | 54.0  | 56.2  | 59.5  | 66.1  | 73.5  | 76.6  | 77.8  | 72.7  | 67.7  | 61.8  | 56.6 |
|                                                                               |             | MCDB   | 58.0  | 57.8  | 60.1  | 64.1  | 71.6  | 78.3  | 81.6  | 82.9  | 78.7  | 72.9  | 66.2  | 60.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.3  | 14.7  | 14.8  | 14.7  | 15.0  | 15.6  | 15.6  | 16.4  | 16.9  | 16.4  | 15.2  | 14.8 |
|                                                                               |             | MCDBR  | 14.5  | 15.5  | 17.7  | 20.1  | 20.6  | 20.7  | 18.8  | 19.3  | 19.1  | 17.7  | 16.3  | 16.0 |
|                                                                               |             | MCWBR  | 10.8  | 10.7  | 11.9  | 11.8  | 11.5  | 12.2  | 11.5  | 12.0  | 11.3  | 11.0  | 11.0  | 11.5 |
|                                                                               | 5% WB       | MCDBR  | 12.6  | 13.6  | 15.2  | 15.5  | 16.5  | 16.3  | 15.4  | 16.5  | 16.7  | 15.5  | 14.2  | 13.6 |
|                                                                               |             | MCWBR  | 10.4  | 10.4  | 11.3  | 11.0  | 10.9  | 11.9  | 11.1  | 12.0  | 11.7  | 11.2  | 10.7  | 10.9 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.352  | 0.375 | 0.416 | 0.459 | 0.455 | 0.437 | 0.434 | 0.441 | 0.441 | 0.413 | 0.382 | 0.356 |      |
|                                                                               | taud        | 2.434  | 2.345 | 2.222 | 2.113 | 2.178 | 2.278 | 2.301 | 2.280 | 2.273 | 2.325 | 2.373 | 2.427 |      |
|                                                                               | Ebn at Noon | 258    | 266   | 266   | 260   | 264   | 268   | 268   | 263   | 255   | 249   | 243   | 246   |      |
|                                                                               | Edh at Noon | 27     | 33    | 41    | 49    | 47    | 42    | 41    | 41    | 39    | 33    | 28    | 25    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 645    | 882   | 1288  | 1673  | 2019  | 2277  | 2333  | 2071  | 1527  | 1085  | 721   | 576   |      |
|                                                                               | RadStd      | 63     | 85    | 115   | 120   | 108   | 131   | 92    | 107   | 80    | 93    | 60    | 60    |      |

## Historical Trends

|                        |       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        | (d)   | (e)   | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)   | (m)   |       |
| Station Only           | N/A   | N/A   | N/A     | N/A    | -1.04   | -1.78 | N/A   | N/A         | N/A   | N/A   |       |
| Regional (7 neighbors) | +0.49 | N/A   | N/A     | N/A    | -0.74   | -1.19 | N/A   | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page